Tasnuba Yasmin

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EDUCATION

Military Institute of Science and Technology (MIST), Dhaka **February 2019 – June2023**
Bachelor of Science**,** Biomedical Engineering **CGPA**: 3.61/4.00 (160 credit hours)

PROFESSIONAL EXPERIENCE

Internship Trainee **June 2024 – Present**
Child Health Research Foundation (CHRF) Dhaka, Bangladesh

- Training on "Laboratory Techniques on Infectious Disease (LTID)" which includes:
  - Pathogen identification: Aseptic sample collection from blood and CSF
  - Antimicrobial susceptibility testing and data analysis for diagnostics
  - Specimen storage and bio-banking
- Training on "Molecular Biology" which includes:
  - DNA & RNA extraction from clinical specimen
  - Primer design for PCRs, Conventional PCR, q-RT-PCR
  - Gel electrophoresis, imaging & analysis
  - Laboratory safety protocols and waste disposal

Research Assistant **May 2023 – February 2024**
Cell and Tissue Engineering Lab, Biomedical Engineering Department, MIST Dhaka, Bangladesh

- **Project**: Evaluation of Mechanical Strength, Chemical Composition, and Antibacterial Activity of Nigella Sativa Extraction-Based Hydrogel.
- **Role:** Hydrogel/Agar plate preparation, Bacterial stocking and culture, antibacterial activity measurement, physical tests of hydrogel like moisture retention, water-vapor testing etc.
- **Key findings:**
  - **Mechanical Strength:** Higher Nigella Sativa conc. enhances tensile strength and rigidity.
  - **Antibacterial Activity**: Pure thymoquinone requires high concentrations for bacterial inhibition but is more effective when combined with other antibacterial compounds.
- **Skill:** Solution preparation and mixing, sterilization techniques, gelation control, physical property testing, incubation methods, preparing and using agar diffusion method (disc diffusion method), quantifying bacterial inhibition zones, analytical techniques, data analysis

Research Intern **October 2023 – December 2023**
Center for Health Innovation, Research and Learning (CHIRAL) Dhaka, Bangladesh
Attended workshops on:

- **"Easy Statistics for Biomedical Researchers"** covering statistical modeling, visualization, sample size determination, data analysis, and common errors.
- **Bioinformatics Bootcamp & Cancer Bioinformatics** focusing on genome sequencing (proteomics, transcriptomics, metabolomics) and database management.
- **"Fundamentals of PCR and PCR Data Analysis"** discussing different PCR types (high-fidelity, colony PCR) and analysis techniques.

Industrial Trainee **February 2022 – March 2022**
Combined Military Hospital (CMH) Dhaka, Bangladesh

- Learned about medical devices maintenance, repairing and hospital management.
- Received Hands on experience with biomedical devices: X-ray, CT scan, ultrasound, MRI etc.

RESEARCH SKILLS

Wet Lab Skills: Antibacterial activity measurement, Bacterial Stocking & culture, Preparation of Liquid media (LB broth) and solid media (LB agar), Disc diffusion method, Zone of inhibition analysis, Nucleic acid extraction, PCR Primer design, Gel electrophoresis, imaging & analysis.
Tools: Magnetic hotplate, Scanning Electron Microscope (SEM), Universal Testing Machine (UTM), Laminar Flow Hood.

Dry Lab Skills: Biomedical Signal and Image Processing, Medical Image segmentation, 3d modeling, Computational modeling, Finite element analysis, Programming, Circuit design and simulation, Motion Analysis.
Software: Ansys (Fluent, Static Structural), MIMICS, SolidWorks, MATLAB, Simulink, Opensim, Kinovea, Python, Microsoft Office, Arduino.

RESEARCH EXPERIENCE

"Development of Garlic-Extract Incorporated PVA/Gelatin-Based Hydrogel For Biomedical Applications."(Undergraduate Thesis) April 2022 – February 2023

Outcome: Garlic juice-based hydrogel had the highest antibacterial activity and endured friction like human skin. Its swelling and moisture retention capability make it ideal for wound dressings.
Skill: Solution formulation, gelation regulation, disk diffusion method, moisture retention assessment, swelling behavior analysis.

"A smart and automatic wheelchair for paralyzed patient" August 2022-October 2022

Outcome: Enhanced mobility and control for paraplegic and quadriplegic patients achieved through hybrid, voice, wrist, and head tilt controls, replacing traditional joystick for improved accessibility and independence. **Skill:** python

"Design and analysis of a silicon based Toe-prosthesis" March 2022-April 2022

Outcome: Created a 3D toe prosthesis model using Solidworks, validated via static structural simulation to optimize load distribution for standing, ensuring improved comfort and functionality.
Skill: Proficiency in 3D modeling with Solidworks and static structural simulation for load distribution analysis, design optimization for prosthetic devices.

SCHOLARLY CONTRIBUTIONS

Yasmin, T, Fariha, A, Hasan, M.M (2023, May), "Development of Garlic-Extract Based Hydrogel For Biomedical Applications" –

- **Abstract Presentation**- "World Conference on Applied Science, Engineering & Technology (WCASET-23)" by IFERP (Institute for Educational Research and Publication), Malaysia.
- **Poster Presentation**-"International Conference on Pharmaceutical Sciences-2023" organized by State University of Bangladesh and the abstract of the research work was published.

ACHIEVEMENTS

- Best paper presentation Award at "44th World Conference on Applied Science, Engineering & Technology (WCASET-23)"
- Dean's honor list (Freshman year) & Merit Scholarship (Spring '20, Fall '20)
- Government Scholarship (Talent pool) in Junior School Certificate Exam.
- "Perfect Attendance Certificate for 100% attendance" (Junior and Senior year of High school)

EXTRA-CURRICULAR ACTIVITIES

- Obtained 2nd prize in a calligraphy competition arranged by **"MIST Literature Club".**
- Volunteered in the national event **"Anuronon-2022"** organized by "Einthoven Club" (MIST)
- Volunteer at **Blind Education and Rehabilitation Development Organization** (**BERDO**)